

Final Presentation

Apples and pears? Impacts of using the Jacobian scaled conditional Akaike Information Criteria for model selection in linear mixed models. Advanced Small Area Estimation (Prof. Schmid)

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Motivation

- ▶ Situation: Linear Mixed Model with $y_i = X_i\beta + Z_iu_i + \epsilon_i$
- ▶ Adequate set of variables essential for good model fit
- ▶ Data driven transformation can improve the model fit
- ▶ Transformations especially important if normality assumption(s)
 $u_i \sim N(0, G)$ and/or $\epsilon_i \sim N(0, \sigma^2 I_n)$ not fulfilled
- ▶ Problem: Variable selection and estimation of transformation parameter λ affect each other:
 - ▶ Optimal set of covariates depends on transformation chosen and vice versa
 - ▶ Other papers suggest the usage of a conditional Akaike Information Criteria: $cAIC = -2\log(g(y|\hat{\theta}, \hat{u})) + 2K_{MLE}$
 - ▶ Problem: Not scale invariant
 - ▶ Naïve solution: variable selection on original scale and transformation afterwards

Proposed Solution I

- ▶ Idea: Change of variable approach:
 $l(y|\theta, u) = g(\tilde{y}|\theta, u) \cdot J(\lambda, y)$, with $l(y|\theta, u)$ being the conditional model density $\rightarrow$ calculation of K-L-divergence between conditional true and model density
- ▶ Solution: Scaling the $cAIC$ with the Jacobian J :
 $JcAIC = -2\log(g(\tilde{y}|\hat{\theta}, \hat{u})) - 2\log(J(\hat{\lambda}, y)) + 2BC$
- ▶ $\tilde{y}$ being the vector of transformed y with transformation $T_{\lambda}(y)$
- ▶ g being the density function of the approximating model
- ▶ J being the Jacobian matrix:
 $J(\hat{\lambda}, y) = \prod_{i=1}^D \prod_{j=1}^{N_i} \frac{\partial \tilde{y}_{ij}}{\partial y_{ij}} = \prod_{i=1}^D \prod_{j=1}^{N_i} (y_{ij} + s)^{\lambda-1}$
- ▶ BC being a bias correction term, estimated by a bootstrap

Proposed Solution II

Consider a Box-Cox transformation ($i = 1, \dots, D$ and $j = 1, \dots, N_i$):

$$T_{\lambda}(y_{ij}) = \tilde{y} = \begin{cases} \frac{(y_{ij}+s)^{\lambda}-1}{\lambda} & \text{if } \lambda \neq 0 \\ \log(y_{ij} + s) & \text{if } \lambda = 0 \end{cases}$$

1. Start with full model with all p variables and estimate $\hat{\lambda}$ for a Box-Cox transformation
2. For each step $s = 1, \dots, S$:
 - i) Consider all possible models which exclude one predictor
 - ii) Estimate $\hat{\lambda}$ for each candidate and calculate $JcAIC$ with estimated $\hat{\lambda}$
 - iii) Compare $JcAIC$ values and choose model with lowest value as new optimal model
3. Compare $JcAIC$ from 2.iii) with the previously optimal model from step $s - 1$. Repeat until no better $JcAIC$ can be found.

Methods

- ▶ Research question: Does the proposed simultaneous selection and estimation outperform the naive selection in different data scenarios?
- ▶ Approach I: Simulation Study
 - ▶ Four different data generation processes
 - ▶ Four different variable selection mechanisms
 - ▶ Inclusion of two irrelevant variables in the model
- ▶ Approach II: Application to real world data
 - ▶ Data from Mozambique
 - ▶ Estimation of head count ration and poverty gap based on naive approach vs. proposed solution

Variable Selection Mechanisms

- ▶ “No”: $T_\lambda(y_{ij}) = y$ with simple stepwise variable selection on original scale
- ▶ “Log”: $T_\lambda(y_{ij}) = \log(y)$ with simple stepwise variable selection on original scale
- ▶ “Box-Cox Naïve”: First stepwise selection on original scale and Box-Cox transformation afterwards
- ▶ “Box-Cox Optimal”: Stepwise variable selection using $JcAIC$ on Box-Cox transformed y

Data generating processes

- ▶ $x_{1,ij} \sim N(\mu_i, 3^2)$ with $\mu_i \sim U[-3, 3]$,
 $x_{2,ij} \sim \text{Bin}(1, 0.8)$, $x_{3,ij} \sim N(0, 1)$
- ▶ Additional irrelevant variables: $z_1 \sim N(1, 0.1^2)$ and
 $z_2 \sim \text{Bin}(1, 0.4)$
- ▶ Normal (Low): $400 - 10x_{1,ij} + 100x_{2,ij} - 10x_{3,ij} + u_i + e_{ij}$,
 $u_i \sim N(0, 30^2)$, $e_{ij} \sim N(0, 60^2)$
- ▶ Normal (High): $400 - 10x_{1,ij} + 100x_{2,ij} - 10x_{3,ij} + u_i + e_{ij}$,
 $u_i \sim N(0, 10^2)$, $e_{ij} \sim N(0, 20^2)$
- ▶ Log: $\exp\{10 - x_{1,ij} + x_{2,ij} - 0.5x_{3,ij} + u_i + e_{ij}\}$
- ▶ Box-Cox:
 $[(10 - x_{1,ij} + x_{2,ij} - 0.5x_{3,ij} + u_i + e_{ij}) \cdot (-0.5) + 1]^{\frac{1}{-0.5}}$

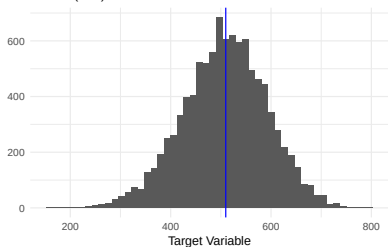
B = 50 vs. B = 200

dgp	transformation	mean_JcAIC_b50	mean_JcAIC_b200	mean_lambda_b50	mean_lambda_b200	proportion_correct_b50	proportion_correct_b200
boxcox	box.cox	-2785.	-2770.	-0.483	-0.483	0.8	0.9
boxcox	boxcox_naive	-2667.	-2638.	-0.490	-0.489	0.3	0.3
boxcox	log	-2020.	-2010.	NaN	NaN	1.0	1.0
boxcox	no	6470.	6480.	NaN	NaN	0.3	0.3
log	box.cox	14838.	14839.	0.00200	0.00197	0.6	0.8
log	boxcox_naive	14914.	14906.	0.00345	0.00336	0.2	0.2
log	log	14837.	14838.	NaN	NaN	0.6	0.8
log	no	21501.	21511.	NaN	NaN	0.2	0.2
normal_high	box.cox	7046.	7040.	1.26	1.35	0.6	0.6
normal_high	boxcox_naive	7098.	7065.	0.951	0.951	0.8	0.8
normal_high	log	7161.	7173.	NaN	NaN	0.8	0.8
normal_high	no	7126.	7137.	NaN	NaN	0.8	0.8
normal_low	box.cox	8352.	8405.	0.963	0.939	0.3	0.3
normal_low	boxcox_naive	8374.	8410.	0.939	0.939	0.7	0.7
normal_low	log	8481.	8492.	NaN	NaN	0.8	0.7
normal_low	no	8451.	8463.	NaN	NaN	0.7	0.7

Table 1: Comparison of results across $B = 50$ and $B = 200$ replications for the Bootstrap of the BC-Term

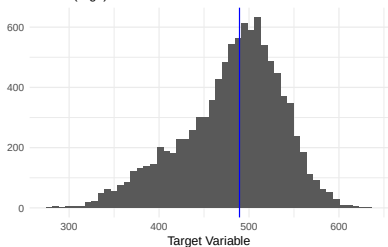
Data generating processes

Normal (Low)



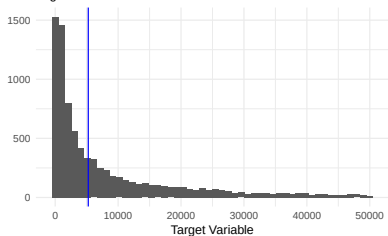
Blue line indicates median

Normal (High)



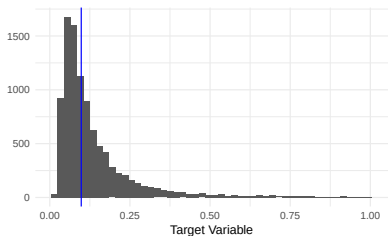
Blue line indicates median

Log



Data truncated due to visibility, $y < 50,000$;
blue line indicates median

Box-Cox

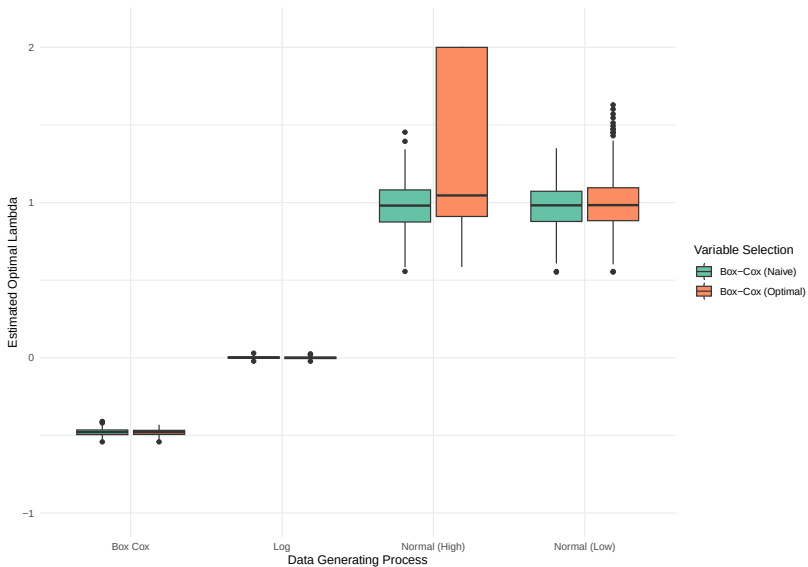


Data truncated due to visibility, $y < 1$;
blue line indicates median

Simulation Setup

- ▶ $R = 300$ iterations
- ▶ $N = 10000$ observations in $D = 50$ domains, sampled via random sample ($n_d \sim Unif(0, 30)$)
- ▶ Population-generation and random sampling based on sample size in each iteration, but same for all methods
- ▶ Interval for estimating λ : $\hat{\lambda} \in \{-2, 2\}$
- ▶ Bias correction term estimated via bootstrap with $B = 50$ (paper: $B = 200$, slight differences as will be discussed in the term paper)

Results: Optimal Lambda's

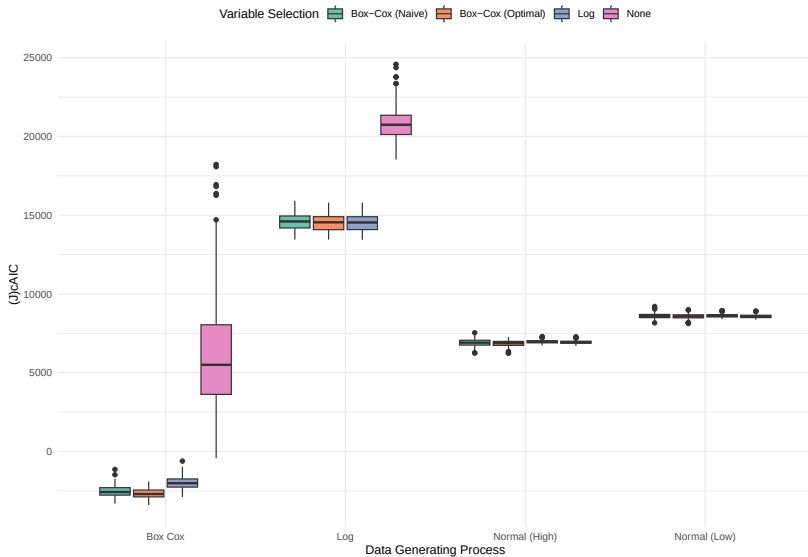


Results: Optimal Lambda's

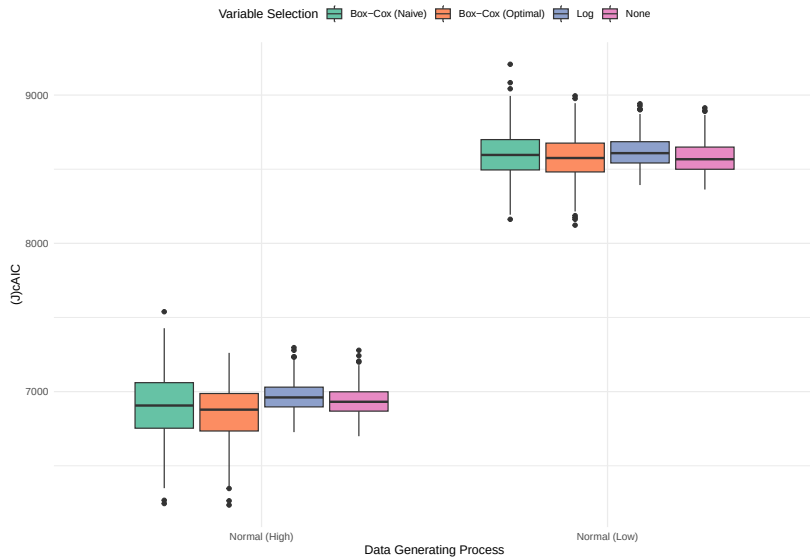
- Expectation: $\hat{\lambda} \approx 1$ for Normal settings $\hat{\lambda} \approx 0$ for Log Setting, and $\hat{\lambda} \approx -0.5$ for Box-Cox-Setting

DGP	Transformation	Min	Mean	Median	Max
Box Cox	Box-Cox (Naive)	-0.54	-0.48	-0.48	-0.41
Box Cox	Box-Cox (Optimal)	-0.54	-0.48	-0.48	-0.43
Log	Box-Cox (Naive)	-0.02	0.00	0.00	0.03
Log	Box-Cox (Optimal)	-0.02	0.00	0.00	0.03
Normal (High)	Box-Cox (Naive)	0.56	0.98	0.98	1.45
Normal (High)	Box-Cox (Optimal)	0.59	1.25	1.05	2.00
Normal (Low)	Box-Cox (Naive)	0.55	0.98	0.98	1.35
Normal (Low)	Box-Cox (Optimal)	0.55	1.01	0.98	1.63

Results: JcAIC



Results: JcAIC - zoomed into Normal Settings



Results: JcAIC

► Normal (Low):

Transformation	Min	Mean	Median	Max
Box-Cox (Naive)	8162.11	8599.32	8596.08	9207.71
Box-Cox (Optimal)	8122.63	8575.04	8575.46	8994.66
Log	8393.09	8617.99	8608.22	8941.14
None	8363.01	8578.25	8567.24	8913.73

► Normal (High):

Transformation	Min	Mean	Median	Max
Box-Cox (Naive)	6245.02	6901.62	6906.25	7538.73
Box-Cox (Optimal)	6235.06	6855.44	6878.43	7261.35
Log	6726.79	6963.73	6960.81	7296.72
None	6699.21	6935.16	6931.43	7278.76

Results: JcAIC

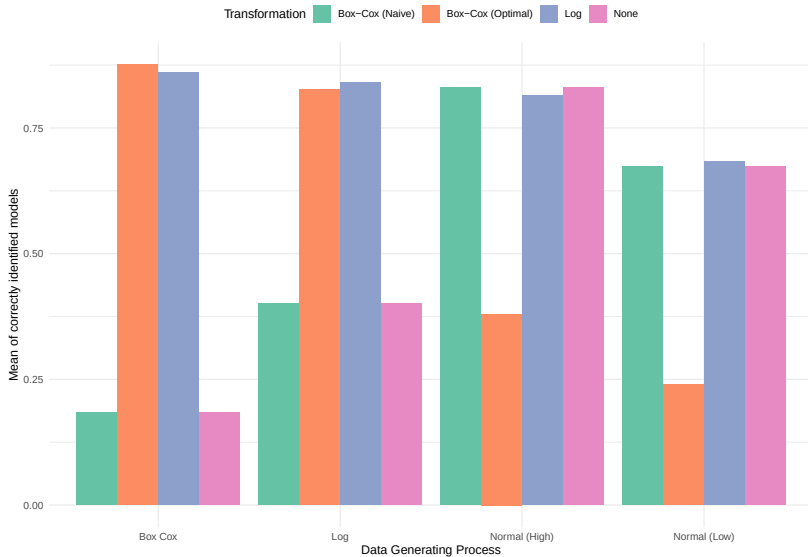
► Log:

Transformation	Min	Mean	Median	Max
Box-Cox (Naive)	13455.08	14572.79	14602.64	15916.29
Box-Cox (Optimal)	13454.51	14508.68	14548.80	15798.44
Log	13442.66	14508.01	14535.83	15802.67
None	18531.23	20803.71	20743.94	24583.03

► Box-Cox:

Transformation	Min	Mean	Median	Max
Box-Cox (Naive)	-3305.61	-2542.10	-2577.17	-1139.78
Box-Cox (Optimal)	-3402.87	-2667.26	-2703.47	-1905.03
Log	-2894.39	-1980.86	-2015.43	-611.39
None	-428.27	6183.83	5500.12	18223.78

Results: Correctly identified variables



Results: Correctly identified variables

► Normal (Low)

Transformation	x1	x2	x3	z1	z2	Correct
Box-Cox (Naive)	1.00	1.00	0.78	0.14	0.14	0.67
Box-Cox (Optimal)	0.82	0.93	0.59	0.39	0.39	0.24
Log	1.00	1.00	0.79	0.14	0.14	0.68
None	1.00	1.00	0.78	0.14	0.14	0.67

► Normal (High)

Transformation	x1	x2	x3	z1	z2	Correct
Box-Cox (Naive)	1.00	1.00	0.99	0.16	0.16	0.83
Box-Cox (Optimal)	0.99	0.73	0.83	0.40	0.40	0.38
Log	1.00	1.00	0.99	0.17	0.17	0.82
None	1.00	1.00	0.99	0.16	0.16	0.83

Results: Correctly identified variables

► Log

Transformation	x1	x2	x3	z1	z2	Correct
Box-Cox (Naive)	1	0.55	0.84	0.16	0.16	0.40
Box-Cox (Optimal)	1	1.00	0.99	0.16	0.16	0.83
Log	1	1.00	0.99	0.15	0.15	0.84
None	1	0.55	0.84	0.16	0.16	0.40

► Box-Cox

Transformation	x1	x2	x3	z1	z2	Correct
Box-Cox (Naive)	0.99	0.53	0.42	0.16	0.16	0.18
Box-Cox (Optimal)	1.00	0.99	1.00	0.11	0.11	0.88
Log	1.00	1.00	1.00	0.14	0.14	0.86
None	0.99	0.53	0.42	0.16	0.16	0.18

Discussion

- ▶ More computational intensive variable selection using the *JcAIC* mostly improves performance compared to naïve methods
 - ▶ Estimated $\hat{\lambda}$ close to true λ , but same/worse performance than naïve method
 - ▶ Same performance for Normal and Log setup and best performance for Box-Cox DGP
 - ▶ Moderate performance regarding variable selection for Normal settings, good for Log and Box-Cox
- ▶ Is it worth the trouble? Yes, if you have the computational power

Thank you for your attention!

Literature

Krennmair, P., Würz, N., Schmid, T., & Tzavidis, N. (2026). Random forests and mixed effects random forests for small area estimation of general parameters: A poverty mapping case study in mozambique. *Annals of Applied Statistics*, 20(1), 809–832. <https://doi.org/10.1214/25-AOAS2126>

Lee, Y., Rojas-Perilla, N., Runge, M., & Schmid, T. (2023). Variable selection using conditional AIC for linear mixed models with data-driven transformations. *Statistics and Computing*, 33(1), 27. <https://doi.org/10.1007/s11222-022-10198-9>